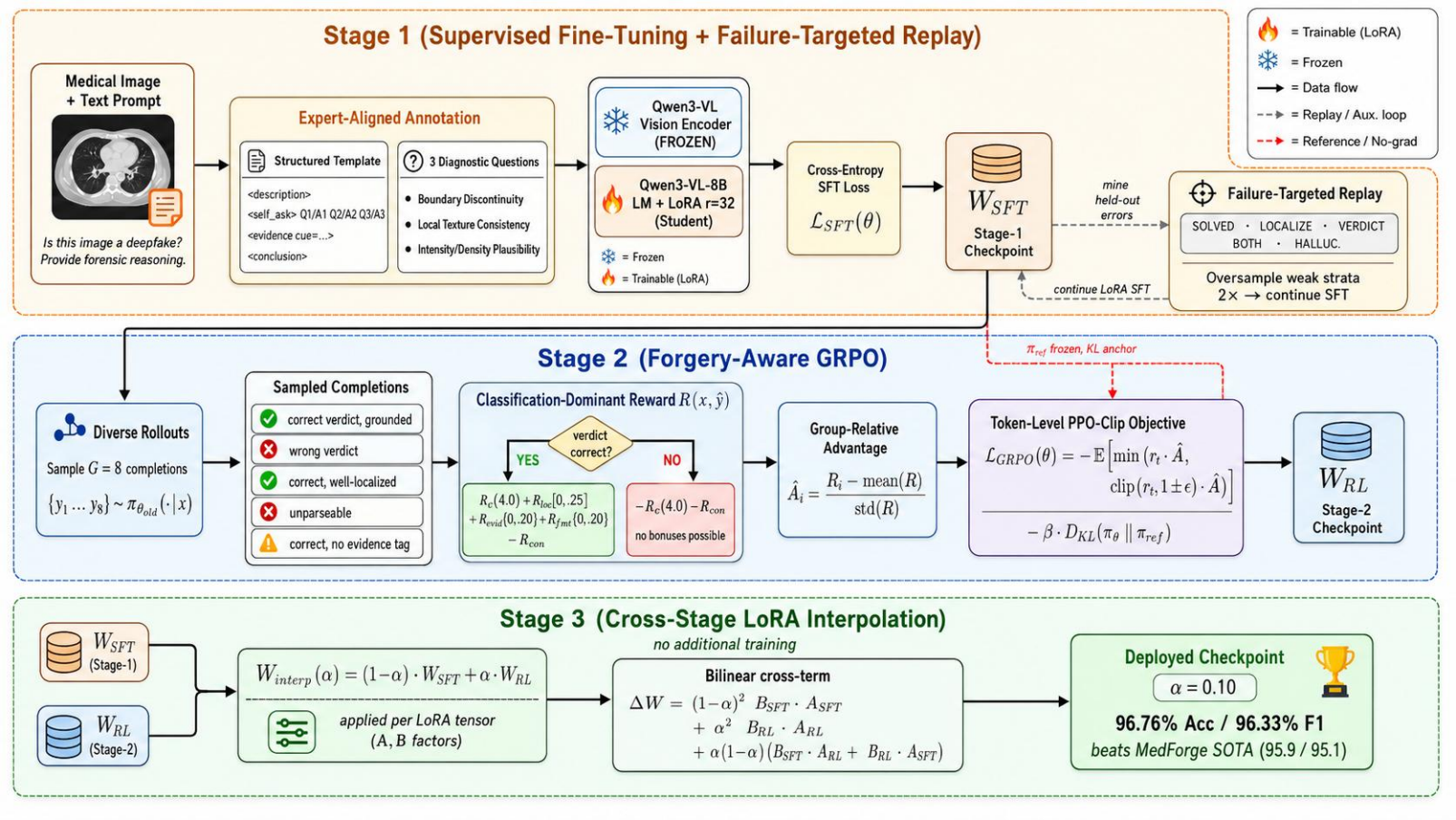

Stage 1 (Supervised Fine-Tuning + Failure-Targeted Replay)
= Trainable (LoRA)
= Frozen
= Data flow
= Replay / Aux. loop
= Reference / No-grad
Medical Image + Text Prompt
Is this image a deepfake? Provide forensic reasoning.
Expert-Aligned Annotation
Structured Template
<description>
<self_ask> Q1/A1 Q2/A2 Q3/A3
<evidence cue=...>
<conclusion>
3 Diagnostic Questions
Boundary Discontinuity
Local Texture Consistency
Intensity/Density Plausibility
Qwen3-VL Vision Encoder (FROZEN)
Qwen3-VL-8B LM + LoRA r=32 (Student)
= Frozen
= Trainable (LoRA)
Cross-Entropy SFT Loss
$\mathcal{L}_{SFT}(\theta)$
W_{SFT}
Stage-1 Checkpoint
mine held-out errors
continue LoRA SFT
Failure-Targeted Replay
SOLVED · LOCALIZE · VERDICT
BOTH · HALLUC.
Oversample weak strata 2× → continue SFT
Stage 2 (Forgery-Aware GRPO)
π_{ref} frozen, KL anchor
Diverse Rollouts
Sample $G = 8$ completions
$\{y_1 \dots y_8\} \sim \pi_{\theta_{old}}(\cdot|x)$
Sampled Completions
correct verdict, grounded
wrong verdict
correct, well-localized
unparseable
correct, no evidence tag
Classification-Dominant Reward $R(x, \hat{y})$
verdict correct?
YES
NO
$R_c(4.0) + R_{loc}[0,.25] + R_{evid}\{0,.20\} + R_{fmt}\{0,.20\} - R_{con}$
$-R_c(4.0) - R_{con}$
no bonuses possible
Group-Relative Advantage
$\hat{A}_i = \frac{R_i - \text{mean}(R)}{\text{std}(R)}$
Token-Level PPO-Clip Objective
$\mathcal{L}_{GRPO}(\theta) = -\mathbb{E}\left[\min\left(r_t \cdot \hat{A}, \text{clip}(r_t, 1 \pm \epsilon) \cdot \hat{A}\right)\right] - \beta \cdot D_{KL}(\pi_\theta \,\|\, \pi_{ref})$
W_{RL}
Stage-2 Checkpoint
Stage 3 (Cross-Stage LoRA Interpolation)
no additional training
W_{SFT} (Stage-1)
W_{RL} (Stage-2)
$W_{interp}(\alpha) = (1-\alpha) \cdot W_{SFT} + \alpha \cdot W_{RL}$
applied per LoRA tensor (A, B factors)
Bilinear cross-term
$\Delta W = (1-\alpha)^2 \; B_{SFT} \cdot A_{SFT} + \alpha^2 \; B_{RL} \cdot A_{RL} + \alpha(1-\alpha)(B_{SFT} \cdot A_{RL} + B_{RL} \cdot A_{SFT})$
Deployed Checkpoint
$\alpha = 0.10$
96.76% Acc / 96.33% F1
beats MedForge SOTA (95.9 / 95.1)